

MALWARE FUNDAMENTALS

1. INTRODUCTION

Malware stands for **Malicious Software**. It refers to software intentionally designed to damage systems, steal information, disrupt operations, gain unauthorized access, or perform other harmful activities.

Malware can affect computers, smartphones, servers, networks, and other connected devices. It can enter a system through malicious email attachments, infected websites, unsafe downloads, compromised software, USB devices, or other methods.

The major types of malware include **viruses, worms, trojans, spyware, ransomware, adware, and rootkits**.

2. OBJECTIVE

The main objective of this task is to understand malware classification and identify the characteristics and effects of different types of malware.

Specific Objectives

- Understand the meaning of malware.
 - Identify major malware categories.
 - Understand how malware spreads.
 - Analyze the effects of malware.
 - Learn basic malware prevention methods.
 - Differentiate between viruses, worms, trojans, spyware, and ransomware.
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3. WHAT IS MALWARE?

Malware is any software or program created with the intention of performing malicious activities.

Depending on its type, malware can:

- Steal sensitive information
- Damage or delete files
- Encrypt data
- Monitor user activity

- Gain unauthorized access
 - Slow down systems
 - Install additional malicious software
 - Disrupt computer operations
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4. MAJOR TYPES OF MALWARE

4.1 VIRUS

A computer virus is malicious code that attaches itself to a legitimate file or program and spreads when the infected file is executed.

Characteristics

- Usually requires user action to execute.
- Can modify or damage files.
- Can replicate by infecting other files.

Effects

- File corruption
- Data loss
- System instability
- Reduced performance

Prevention

- Use security software.
 - Avoid suspicious files and downloads.
 - Keep software updated.
 - Scan external storage devices.
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5. WORM

A worm is malware that can **self-replicate and spread across systems or networks without requiring a user to manually execute an infected file each time.**

Characteristics

- Can spread automatically.

- Often exploits security vulnerabilities.
- Can spread rapidly across networks.

Effects

- Network congestion
- System slowdown
- Resource consumption
- Disruption of services

Prevention

- Keep operating systems updated.
 - Apply security patches.
 - Use firewalls and network security controls.
 - Monitor unusual network activity.
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6. TROJAN

A Trojan is malware that disguises itself as legitimate or useful software to trick users into installing or running it.

Unlike a virus or worm, a Trojan generally does not self-replicate.

Example

A user downloads an application that appears to be legitimate, but the application secretly installs malicious software.

Effects

- Unauthorized access
- Data theft
- Installation of additional malware
- System compromise

Prevention

- Download applications only from trusted sources.
- Avoid pirated or unknown software.
- Verify software before installation.
- Keep security software updated.

7. SPYWARE

Spyware is malware designed to secretly monitor a user's activities and collect information without proper authorization.

It may collect:

- Browsing activity
- Personal information
- Login credentials
- System information
- Other sensitive data

Effects

- Privacy loss
- Credential theft
- Data leakage
- Unauthorized monitoring

Prevention

- Install applications from trusted sources.
- Use reputable security software.
- Keep systems updated.
- Avoid suspicious links and downloads.

8. RANSOMWARE

Ransomware is malware that prevents users from accessing their files or systems, commonly by encrypting data, and demands payment from the victim.

Example

An employee opens a malicious attachment. The ransomware encrypts company files and displays a message demanding payment.

Effects

- Files become inaccessible.
- Business operations may stop.

- Financial losses may occur.
- Sensitive information may be exposed.

Prevention

- Maintain regular backups.
 - Keep software updated.
 - Avoid suspicious attachments and links.
 - Use security controls and endpoint protection.
 - Provide cybersecurity awareness training.
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9. ADWARE

Adware is software that displays unwanted advertisements. Some forms of adware may also track user activity or redirect users to unwanted websites.

Effects

- Unwanted advertisements
- Browser redirects
- Reduced system performance
- Possible privacy concerns

Prevention

- Download software from trusted sources.
 - Avoid suspicious freeware and websites.
 - Review applications before installing them.
 - Use appropriate security software.
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10. ROOTKIT

A rootkit is malware designed to hide malicious activity and maintain unauthorized access to a system.

It can be difficult to detect because it may attempt to conceal files, processes, or other malicious activity.

Effects

- Hidden unauthorized access

- System compromise
- Data theft
- Difficult malware detection

Prevention

- Keep systems updated.
- Use trusted security software.
- Restrict administrative privileges.
- Monitor systems for unusual activity.

11. MALWARE COMPARISON TABLE

Malware Type	Main Characteristic	Typical Effect
Virus	Attaches to files and spreads when executed	File corruption and system damage
Worm	Self-replicates and spreads across networks	Network disruption and resource consumption
Trojan	Disguises itself as legitimate software	Unauthorized access and data theft
Spyware	Secretly monitors user activity	Privacy loss and information theft
Ransomware	Encrypts or blocks access to data	Data loss and financial disruption
Adware	Displays unwanted advertisements	Unwanted ads and reduced performance
Rootkit	Hides malicious activity and maintains access	Difficult detection and system compromise

12. VIRUS VS WORM

Feature	Virus	Worm
Replication	Usually requires a host file/program	Can self-replicate
User Interaction	Often requires execution	May spread automatically
Main Spread	Infected files/programs	Networks and system vulnerabilities
Main Impact	File/system damage	Network and system disruption

13. TROJAN VS VIRUS

Feature	Trojan	Virus
Disguise	Usually disguises itself as legitimate software	Attaches to files/programs
Self-Replication	Generally does not self-replicate	Can replicate by infecting files
Main Purpose	Gain access or deliver malicious payloads	Spread and infect files
User Action	Often requires user installation/execution	Usually requires execution of infected file

14. EFFECTS OF MALWARE

Malware can have several negative effects on individuals and organizations.

1. Data Theft

Sensitive information may be stolen.

2. Data Loss

Files may be deleted or corrupted.

3. Financial Loss

Organizations may lose money due to fraud, ransomware, or business disruption.

4. Privacy Loss

Personal or confidential information may be monitored or exposed.

5. System Damage

Malware can damage files, applications, or operating system components.

6. Service Disruption

Business operations and online services may become unavailable.

7. Unauthorized Access

Attackers may gain access to accounts, systems, or networks.

15. COMMON MALWARE ENTRY METHODS

Malware can enter a system through:

- Phishing emails
 - Malicious attachments
 - Fake websites
 - Unsafe downloads
 - Pirated software
 - Infected USB devices
 - Compromised websites
 - Unpatched software vulnerabilities
 - Malicious advertisements
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16. MALWARE PREVENTION

Users and organizations can reduce malware risks by following these practices:

1. Keep operating systems and applications updated.
2. Use reputable security software.
3. Avoid suspicious links and attachments.
4. Download software only from trusted sources.
5. Maintain regular backups.
6. Avoid pirated or unauthorized software.
7. Use strong authentication and MFA.
8. Follow the principle of least privilege.

9. Scan external storage devices.
 10. Provide regular cybersecurity awareness training.
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17. BASIC MALWARE RESPONSE

If malware is suspected:

1. Disconnect the affected device from the network when appropriate.
 2. Do not continue interacting with suspicious files or applications.
 3. Inform the IT or security team.
 4. Run appropriate security checks.
 5. Identify and remove the malicious software using approved procedures.
 6. Change compromised credentials.
 7. Restore affected files from trusted backups if necessary.
 8. Investigate how the infection occurred to prevent recurrence.
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18. CONCLUSION

Malware is a major cybersecurity threat that can affect individuals, businesses, and organizations. Different types of malware have different methods and purposes.

Viruses infect files, **worms** spread automatically across networks, **Trojans** disguise themselves as legitimate software, **spyware** secretly monitors users, and **ransomware** commonly blocks access to data.

Understanding these malware categories helps users identify potential threats and take appropriate preventive measures. Regular software updates, security awareness, trusted downloads, strong authentication, and reliable backups are essential for reducing malware-related risks.